The Evolution of Transit Access to Essential Services

Sierra Arnold*

Draft: November 9, 2026

Click here for the most recent version

Abstract

A household that depends on public transit must reach not only jobs but the bank, the grocery store, the library, the hospital, and the community college. We construct a multi-service panel of neighborhood transit access designed to be differenced and decomposed: archived transit schedules for 52 U.S. metropolitan areas, routed quarterly on one fixed method from 2013 to 2025, with five administrative destination layers scored on the same block-group travel-time matrices as job access. Each service's within-neighborhood change splits exactly into a transit-network component and a destination component in the 19 metros with clean archives. The mechanisms differ sharply by service. Bank access fell 18.1 percent, driven by branch closures; library access fell 2.5 percent, driven by the network; hospital access fell 3.4 percent, split between the two sides; grocery access roughly held (up 1.7 percent) as store growth offset network decline and store shifts toward transit-poor locations; community colleges were flat once reclassified and consolidated colleges are not mistaken for closures. The metro-clustered intervals for the bank and hospital declines exclude zero; the smaller changes describe these metros rather than a national process. Deep losses compound. Neighborhoods losing at least four of six services, jobs included, are 5.5 times as common as independent losses would imply, and those losing five or more are 31.5 times as common. These neighborhoods are poorer, more carless, and more minority. The side of access that moves depends on the service, not on the era.

Keywords: accessibility; public transit; essential services; transport equity; healthcare access; social exclusion

*Xavier University. Contact: arnolds7@xavier.edu. Transit schedules include data from Transitland (https://www.transit.land/terms). Street network data © OpenStreetMap contributors, available under the Open Database License (https://www.openstreetmap.org/copyright). Routing was performed on resources provided by the Ohio Supercomputer Center (Ohio Supercomputer Center, 1987, Columbus, OH, https://ror.org/01apna436), whose support we gratefully acknowledge.

1 Introduction

A household without a car reaches the bank, the grocery store, the library, the hospital, and the community college by transit, the same way it reaches work. Across the 90 analysis metros with census attributes, 7.1 million households (10.0%) have no vehicle; outside New York, 4.9 million (7.6%) do (U.S. Census Bureau, 2023a). Nationally, about 8% of households lack a vehicle, and 19% of lower-income and 21% of Black adults seldom or never drive, against 3% of upper-income and 7% of White adults (Pew Research Center, 2024; U.S. Census Bureau, 2023b). For these households, the facilities they can reach by transit within a fixed travel time (45 minutes here) are the facilities they can practically use.

In urban food deserts, the exit of transit service is associated with fewer grocery trips by lower-income households and less healthy food purchases (Arnold, 2025). Bank branches still matter to households and businesses even as consolidation removed nearly a fifth of them (Adams and Sherlund, 2025). College-going is largely local, with most students enrolling near home, so a community with no college nearby is an education desert (Hillman, 2016). Spatial access to care is the organizing concern of a large healthcare-access literature (Neutens, 2015; Buchalter et al., 2025). A neighborhood that loses access to several of these services at once shows the inaccessibility to essential goods and services that the transport-poverty literature links to social exclusion (Lucas, 2012).

Over 2014–2025 both the destinations and the transit networks changed. Branches consolidated; supermarkets and hospitals opened and closed; and agencies cut service as ridership fell during the pandemic (Kar et al., 2022). When a neighborhood's access to an essential service fell, which changed: the network or the destinations? The distinction matters for policy. A transit agency can do nothing about branch consolidation; a banking regulator can do nothing about a bus-network redesign. A measured access decline is actionable only once it is attributed, and no cross-sectional measurement can attribute it.

Accessibility research has mostly measured destination types one at a time (Section 2), so relatively little is known about whether different essential services evolve in similar or fundamentally different ways over time. This paper asks that question. Using one longitudinal accessibility database, we compare six services across 52 U.S. metropolitan areas with identical measurement throughout: five essential services, plus job access as the anchor. The quarterly panel, routed on one method held fixed over time, and the network-versus-destination decomposition come from the companion paper (Arnold, 2026); this paper applies both to a new question rather than redeveloping them. There are three main results. First, trends differ by service. Bank access fell 18.1% over the panel while grocery access rose 1.7%, with libraries, hospitals, and community colleges in between. Second, the mechanisms differ even

more. An exact decomposition splits each service's change into its transit-network and destination parts in the 19 metros with clean archives. Bank access fell mainly through branch closures, library access through the network, and hospital access through both. Which side of access moves depends on the service rather than the era. Third, losses compound in the deep tail. Neighborhoods losing four or more services are 5.5 times as common as independent losses would imply, and those losing five or more are 31.5 times as common. These neighborhoods are poorer, more carless, more minority, and more central. The claims are descriptive throughout and concern access itself, not its effects on outcomes.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the travel-time matrices and the destination layers, and Section 4 defines the access measure and its decomposition. Section 5 reports each service's trajectory and decomposition. Section 6 compares the six services, identifies which neighborhoods had the largest declines, and tests whether losses compound across services. Section 7 reports New York separately. Section 8 concludes, and Appendix A holds the robustness and validation exhibits.

2 Related literature

This paper draws on three literatures. The first is accessibility measurement. The cumulative-opportunity construct builds on Hansen (1959) and Wachs and Kumagai (1973), and Handy and Niemeier (1997) and Geurs and van Wee (2004) survey its taxonomy and trade-offs. The health-geography literature measures access to care with floating-catchment and gravity constructs (Luo and Wang, 2003), and Cui and Levinson (2020) formalize the primal/dual distinction. Longitudinal measurement exists for jobs. The *Access Across America* series scores transit access to employment annually across roughly fifty metros (Owen et al., 2025), and Foth et al. (2013) track access against social need across two census years in Toronto. Boisjoly and El-Geneidy (2016) find that a constant 8 a.m. measure of job accessibility in Toronto is representative of relative accessibility across the day. Farber et al. (2014) document meaningful within-day variability for transit access to supermarkets specifically. We hold the departure time fixed across the panel (Section 4). Published schedule data also support equity assessment directly. Karner (2018) builds route-level accessibility measures from public feeds to evaluate transit service equity. These studies established repeated measurement, but none differences cleanly over a long panel or attributes changes to their sources. Multi-service measurement also exists. The TransitCenter Equity Dashboard (Klumpenhouwer et al., 2021) tracks routed transit access to healthcare and groceries in seven regions from 2020 on, and Li et al. (2022) compute one-date 2SFCA indices for several

point-of-interest classes in five cities. Kar et al. (2022) compare block-group transit access to groceries and health care across three phases of 2020 in 22 U.S. cities, with destinations fixed at 2019 locations. In their data, block groups where several social disadvantages coincided were more likely to lose access, which is compounding across disadvantages rather than across services (compare Section 6.3).

The second literature concerns the components of access. The conceptual split of access change into transport and land-use parts is in Geurs and van Wee (2004); longitudinal decompositions exist for European road accessibility (Stępniak and Rosik, 2018; Rosik et al., 2025), and Zhang et al. (2026) follow New York City block groups across four snapshots (2014–2022), separating transit-network change from residential sorting in a widening racial access gap. In the U.S., Janatabadi and Ermagun established a framework for untangling the place and mobility components of access at a point in time. In their framework, holding destinations counterfactually uniform isolates the contribution of transit service to spatial inequality. They apply it to employment access in the Washington region and the City of Chicago (Janatabadi and Ermagun, 2024a,b). In our terms, their "mobility" or "transit service" side is the network component and their "place" or "land use" side the destination component. Ermagun and Witlox (2024) established metro-scale comparison across fifty MSAs, and Ermagun et al. (2025) established fine-grained, time-denominated (nearest-destination) access to six destination types across Chicago, showing that which service binds varies sharply across neighborhoods. Their finding that inequality indices are sensitive to the travel-time budget (Janatabadi and Ermagun, 2024b) is one reason we report the full 15/30/45/60-minute curve.

The third is the set of service-domain literatures, each with its own convention. For banking, Adams and Sherlund (2025) measure block-level *distance to the nearest branch* through the consolidation decade and find it moved little; our count-based transit measure falls 18%, and Section 5.1 reconciles the two. For healthcare, Neutens (2015) reviews the accessibility tradition for transport geographers. Buchalter et al. (2025) build a multi-modal composite across 56 provider classes from quarterly 2016–2020 provider files, with transit times routed on feeds from an open feed database. Food access has the USDA food-desert measures (Ver Ploeg et al., 2009) and a transit-specific literature measuring supermarket access within a city as it varies over the day (Farber et al., 2014) and with commuters' space-time constraints (Widener et al., 2015). Education has "education deserts", counts of colleges within commuting zones (Hillman, 2016), and libraries are benchmarked by average distance to the nearest outlet (Donnelly, 2015). On the transit side, Jomehpour Chahar Aman and Smith-Colin (2020) identify transit deserts by overlaying transit supply on transit-dependent demand. We depart from these distance- and area-based conventions (Section 4) and report

nearest-facility levels as context. The motivating frame throughout is the transport-poverty and social-exclusion literature (Lucas, 2012; Pereira et al., 2017) rather than spatial mismatch. Martens (2012) supplies its normative reading, in which accessibility is the transport good whose distribution is at stake. The social-exclusion frame treats disadvantage as multidimensional (Lucas, 2012), so the quantity of interest here is losing several essentials at once, not any single service. The equity questions of Section 6.2 follow the U.S. work on nonwork accessibility by Grengs (2015) and the activity-participation reading of transport poverty in Allen and Farber (2020).

This paper adds three things to these literatures. To longitudinal measurement it adds a differencing design: one fixed method applied to a 2013–2025 archive and corrected for archive-coverage changes. None of the repeated measures reviewed in this section can be differenced in this way. To the components literature it adds a network-versus-destination attribution of change that is comparable across metros and services. To the service-domain literatures it adds six services measured identically, which makes it possible to measure the overlap of losses across services, something a single-domain study cannot do.

3 Data

3.1 Travel-time matrices (shared infrastructure)

The routing panel is inherited unchanged from Arnold (2026): archived GTFS schedules for every covered metro, one sample date per quarter, routed door-to-door between 2020-vintage census block-group centroids on OpenStreetMap networks with R5 (Conway et al., 2018). Departures span the 7:00–9:00 morning-peak window, and each pair's value is the median door-to-door time over departures within it: the initial walk, waiting, in-vehicle time, transfers, and the final walk. Trips longer than 90 minutes are treated as out of reach, and all other routing parameters are engine defaults, held fixed across metros and quarters. Centroid-to-centroid routing introduces travel-time error in large peripheral block groups. The centroids are fixed, so this error is constant over time and nets out of within-neighborhood change. The archive spans 2013–2025; the analysis window, and every quoted change, runs 2014–2025 (48 quarters). Feed-quality flags (outage and thin-feed quarters) carry over; flagged cells are excluded from value exhibits, with flags-included companions beside each trajectory. The balanced panel covers 52 metros outside New York, described in Section 3.3, summarized in Appendix Table A.1, and listed in supplementary Table S1; New York is reported separately throughout because its gradients differ qualitatively (Section 7).

3.2 Destination layers

Each service comes from a public administrative census of facilities (Federal Deposit Insurance Corporation, 2025; Institute of Museum and Library Services, 2026; U.S. Department of Agriculture, Food and Nutrition Administration, 2026; National Center for Education Statistics, 2025; Centers for Medicare and Medicaid Services, 2026), geocoded and aggregated to 2020 block groups, held constant within calendar years so that sub-annual access movement is network-driven by construction. The library and community-college files end in 2024 and are held for 2025. Table 1 summarizes.

Table 1: Destination layers.

Service	Source	Definition (lead)	2014	Latest
Bank branches	FDIC Summary of Deposits	full-service branches	89,673	73,616
Libraries	IMLS Public Library Survey	stationary outlets (CE+BR)	16,906	16,864
Groceries	USDA SNAP retailer history	supermarkets, superstores, large grocery stores	42,110	44,027
Community colleges	NCES IPEDS	persistent-campus public 2-year	1,045	1,065
Hospitals	CMS Provider of Services (Dec. vintages)	durable general acute care (geocoded; 97–98% coverage)	4,694	4,383

National counts of the geocoded lead layers, totals over the fifty states, DC, and the territories each source covers; latest = 2025, or 2024 for libraries and community colleges. In 2023 the 92 analysis metros hold 35–55% of each layer's national facilities (banks 51%, libraries 38%, groceries 55%, community colleges 35%, hospitals 35%), and the 52 panel metros hold 23–36%.

Four definitional choices deserve note. Two follow domain conventions (banks, groceries), and two respond to measured artifacts in the raw administrative categories (community colleges, hospitals). *Banks*: the lead counts full-service offices (96–97% of all offices), as Adams and Sherlund (2025) do; the all-office variant is indistinguishable in trend. *Groceries*: the lead is full-line food retail (supermarkets, superstores, large grocery stores), the USDA food-desert family (Ver Ploeg et al., 2009); all SNAP-authorized retailers, including convenience and dollar stores, form a broader variant (Section 5.3). *Community colleges*: two artifacts both lower the raw count. IPEDS sector codes reclassify community colleges that begin awarding bachelor's degrees (140 of the 1,047 public two-year institutions in the 2013 file had been reclassified by 2024), and consolidations merge multi-campus systems into single records while the campuses keep operating (48 of the 64 institutions that left the file by 2024). The lead therefore assigns each institution a persistent identity and carries a consolidated system's campuses forward on IPEDS's own merger crosswalk (53 campuses across 22 systems), leaving the national campus count essentially flat, 1,045 to 1,065; Section 5.4 quantifies the decline the raw categories would show. *Hospitals*: CMS certified hundreds of temporary facilities during the pandemic (264 new general-acute certifications in 2020

alone, 225 of them gone by December 2023), so the lead counts *durable* facilities: certifications active in at least three consecutive December vintages, with facilities present in 2013 and 2024–25 entrants retained (censoring grace at both edges); 184 of 8,440 certifications are excluded as transient, 125 of them first seen in 2020–21. The rule removes only the shortest-lived of the 2020 cohort (48 of 264); most of the rest enter and leave the durable set (Section 5.5). The healthcare object is hospitals because we found no public site-level history of community health centers before 2021. Lead hospitals resolve to coordinates at 97–98% per year (91% for all hospitals; Appendix Table A.2), against ≥98–99% for the storefront services. The ungeocoded set is persistent rather than churning, so within-neighborhood trends compare the same facilities over time, and levels are understated by roughly the coverage gap. We audit every service's classification codes and reporting units for these artifacts; groceries' re-typing flows are small and symmetric, and banks' and libraries' codes are stable within facility. The grocery layer reconciles against an independent count (Census County Business Patterns; U.S. Census Bureau, 2025), and the library layer against the survey's own system-level files. The hospital layer's state counts correlate at 0.997 with the HIFLD facility layer (Homeland Infrastructure Foundation-Level Data, 2020), which also supplied some of its coordinates, and certified durable hospitals number a median 0.90 of HIFLD's open general acute and critical-access hospitals by state (Appendix Table A.3). The national totals in Table 1 check metro coverage from the destination side. Over the window the national layers moved −17.9% (branches), −0.2% (outlets), +4.6% (full-line stores), +1.9% (campuses), and −6.6% (durable hospitals). Four of the five moved in the same direction as the covered metros' scale components, so those destination trends are not artifacts of metro coverage. Library outlets were flat nationally but rose in the covered metros (scale +3.3 log points in the clean-archive core).

3.3 How representative are the 52 metros?

The balanced panel is the set of metros whose archived schedules support routing in at least 36 of the 48 quarters from 2014 through 2025. This definition reflects archive coverage, not the metros themselves. The 52 balanced metros outside New York hold roughly 45% of U.S. metropolitan population and include 30 of the 50 largest metros. The routed panels cover 96 metros in all. Of these, 92 enter the unbalanced analysis panel (Table S1 of the online supplement), and 90 carry merged census demographic attributes (189 million residents); the two Connecticut metros' attributes did not merge. Several legacy-transit metros fall below the balance bar because their schedules were missing from the archive or unavailable to this build: San Francisco (22 of 48 quarters), Minneapolis (35), Detroit (3), and Baltimore (34).

Washington, DC is present, but its principal operator, WMATA (Metrorail and Metrobus), is missing throughout, so its routed access is understated and we read that metro with caution. Excluding it changes neither of the two trajectories whose metro-clustered intervals exclude zero (banks −18.1% unchanged, hospitals −3.4% to −3.3%, identical block-group-clustered intervals; Appendix Table A.3). Four other metros, Riverside–San Bernardino, Jacksonville, Palm Bay, and Fairbanks, are routed for transit but absent from the analysis panel, because their car-travel routing failed and the panel requires both. Archive coverage favors larger, better-archived agencies. We therefore read the results as describing transit access in these metros, not in the average U.S. metropolitan area.

4 Measuring access and decomposing its change

4.1 Access measure

Access for block group i, service s, quarter t is the count of s-facilities reachable from i within τ minutes by transit and walking, with $\tau = 45$ as the primary budget; the full 15/30/45/60 curve is in Appendix Table A.4, where every declining trajectory grows in magnitude with the budget (groceries stay between +1.3 and +1.8 percent) and the 45-minute column reproduces the estimates reported here. One construct serves as both trajectory and decomposition object for every service, which is what makes the cross-service comparison possible on the reported measure itself. The domain conventions are carried as robustness variants. Cumulative and gravity measures of job access correlate above 0.9 when the cumulative threshold is set at the mean commute time (Santana Palacios and El-Geneidy, 2022). Among the construct variants of Appendix Table A.5, the only one that reverses a trajectory's sign is community-college coverage (+0.5 points against −0.4%, a trajectory that spans zero under metro clustering).

Departure timing deserves its own statement, because it is the one convention this measure cannot vary. Access is routed in the morning peak, while trips to banks, libraries, groceries, and hospitals are not limited to the morning peak, and transit access to supermarkets varies within the day (Farber et al., 2014). Across 22 U.S. cities in 2020, changes in transit access to groceries and health care differed little between peak and off-peak hours (Kar et al., 2022). After 2020, all-day service recovered faster than peak-period job access in the 19 metros of the clean-archive core (Arnold, 2026). The direction of any departure-time bias for these services is therefore not established, and the network components reported here describe morning-peak access only.

All trend estimates are within-neighborhood. We regress log access (plus one) on quarter

effects with block-group-by-archive-regime fixed effects, block-group population weights (the fixed 2018–22 American Community Survey vintage, constant over the panel; U.S. Census Bureau, 2023a), and standard errors clustered by block group. Composition changes and archive-coverage regime changes therefore do not appear as access changes. Quoted endpoints compare the 2025 annual mean of quarter effects to the 2014 annual base, and seasonality is small (Appendix Table A.3).

4.2 Decomposition

For each metro we fix the base network b (the first unflagged 2014 quarter) and base destination vector (2014), and compute access under all four combinations of network era and destination vintage. The observed within-neighborhood log change splits exactly and symmetrically (a two-factor Shapley decomposition):

$$\underbrace{\ln A(n_t, d_t) - \ln A(n_b, d_b)}_{\text{observed}} = \underbrace{\tfrac{1}{2}\big[\Delta_n \mid d_b\big] + \tfrac{1}{2}\big[\Delta_n \mid d_t\big]}_{\text{network}} + \underbrace{\tfrac{1}{2}\big[\Delta_d \mid n_b\big] + \tfrac{1}{2}\big[\Delta_d \mid n_t\big]}_{\text{destination}}, \qquad (1)$$

with the destination part further split into a geometry-free *scale* term and a residual *relocation* term. Scale for metro m in year y is $\ln(T_{my}/T_{m,2014})$, the log change in the metro's total facility count on the fixed block-group universe. Relocation is the destination component minus scale: the part of the destination effect attributable to *where* facilities sit relative to the network rather than how many exist. The term is broader than facilities physically moving; it captures differential closure and entry by location, and Section 5.1 interprets it that way. The jobs anchor used throughout is the companion paper's layer under the identical specification: LODES workplace employment (all jobs; U.S. Census Bureau, 2026; vintages end in 2023 and are held thereafter), scored on the same matrices, decomposed by the same split (Arnold, 2026). Block-group changes are computed first, population-weighted averages second, and the interaction is allocated symmetrically (Shorrocks, 2013). Archive-regime jumps and the sample rule are described in Appendix A. Appendix Figure A.1 shows the correction in one metro, and Appendix Table A.6 lists the quality flags that define the excluded metros. Every result outside the New York section (Section 7) uses the corrected series. Because archive entry adds measured access on net, the uncorrected series is a diagnostic upper bound in aggregate. Excluding the flagged metros changes levels but, except for hospitals (Section 5.5), not mechanisms. Appendix Figure A.2 compares the component splits in the two samples. All access logs are $\ln(A + 1)$, so zero counts enter at zero and proportional changes are compressed for sparse services.

5 The evolution of access, service by service

Unless noted, every result in Sections 5–7 is the cumulative 45-minute count, population-weighted, within-block-group, archive-entry corrected, and quoted as the 2025 annual mean against the 2014 annual base. *Trajectories* run on *the panel*, *decompositions* on *the clean-archive core*, with the all-metro companion reported beside every core decomposition; both samples are defined in Appendix A, and every exhibit names its own (Appendix Table A.7). The two constructions can differ where flagged metros differ from clean ones, and we flag each such divergence where it occurs. Inference has two levels. The block-group-clustered intervals quoted throughout describe the panel, while the metro-clustered intervals reported with each trajectory are the test for any claim about a common national process.

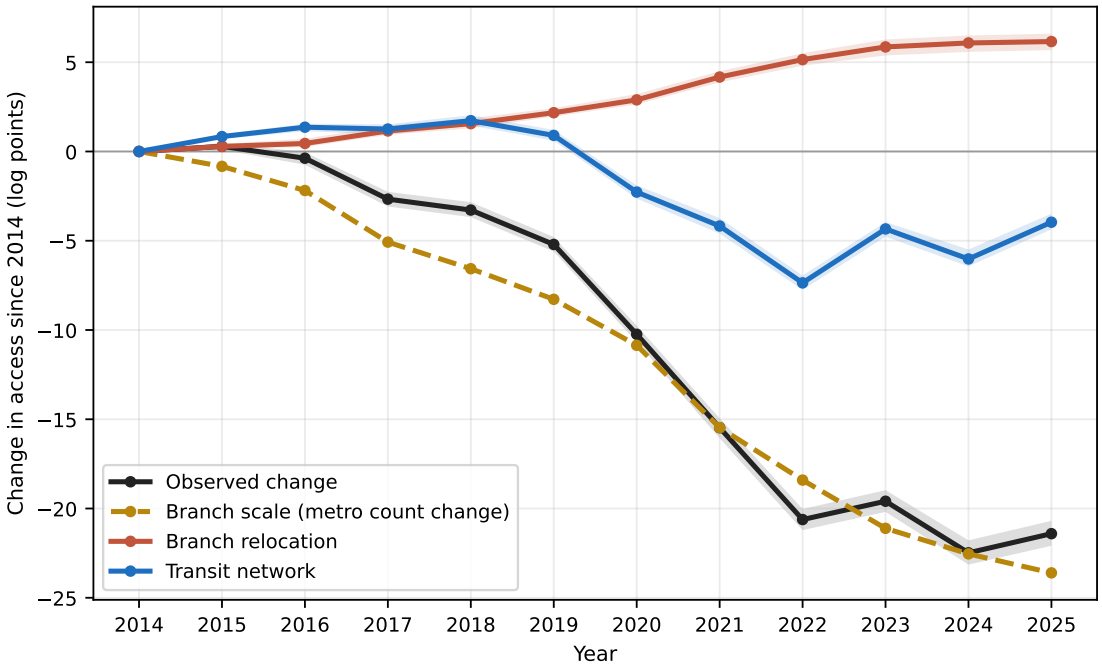
5.1 Banks

Banks are the destination-driven service. Figure 1 decomposes the change over 2014–2025. At the 2025 endpoint the observed −21.4 log-point change splits into −4.0 network, −23.6 branch scale, and +6.2 relocation (all-metro companion: $-17.9 = -3.2 - 19.9 + 5.2$; the mechanism split is the same, and the decline is larger in the clean-archive core). Branch closures, not transit change, removed roughly a fifth of the branches the typical neighborhood could reach within 45 minutes, and the observed line tracks the branch-count line closely.

The *positive* relocation term does not mean that closures avoided accessible places. At the branch level, closure rates were mildly *higher* in the most transit-accessible quartile of 2014 locations than in the least (35.5% against 31.0%, a 4.5-point difference with a 0.8-point standard error; Table 2). The term instead reflects how many branches each neighborhood reached before the closures. Where many branches are reachable, a closure removes one from a large set; where few are, a single closure can leave none within 45 minutes. Net destination change therefore reduced access less than a uniform decline of the same size would have, which is what a positive relocation term measures.

In trajectory terms, transit access to full-service branches fell **18.1%** [95% CI: −18.6, −17.6] between the 2014 and 2025 annual means (flags included: −16.1%; all offices: −18.4%; NYC included: −18.7%). The metro-clustered interval also excludes zero ([−22.5, −13.3]). The decomposition reconciles our finding with Adams and Sherlund (2025). Distance to the *nearest* branch barely changed over the same period (their result; our nearest-facility variant rises by 2 minutes, Appendix Table A.5), while the number of reachable branches fell (our count). The two constructs measure different things, and the decomposition explains why they diverge.

Figure 1: Decomposition of 45-minute transit access to bank branches, 2014–2025.



Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change.

Table 2: Branch closures by 2014 access quartile.

2014 branch location	Branches	Closed by 2025
Q1 (least transit access)	7,191	31.0%
Q2	7,191	31.4%
Q3	7,190	31.3%
Q4 (most transit access)	7,191	35.5%

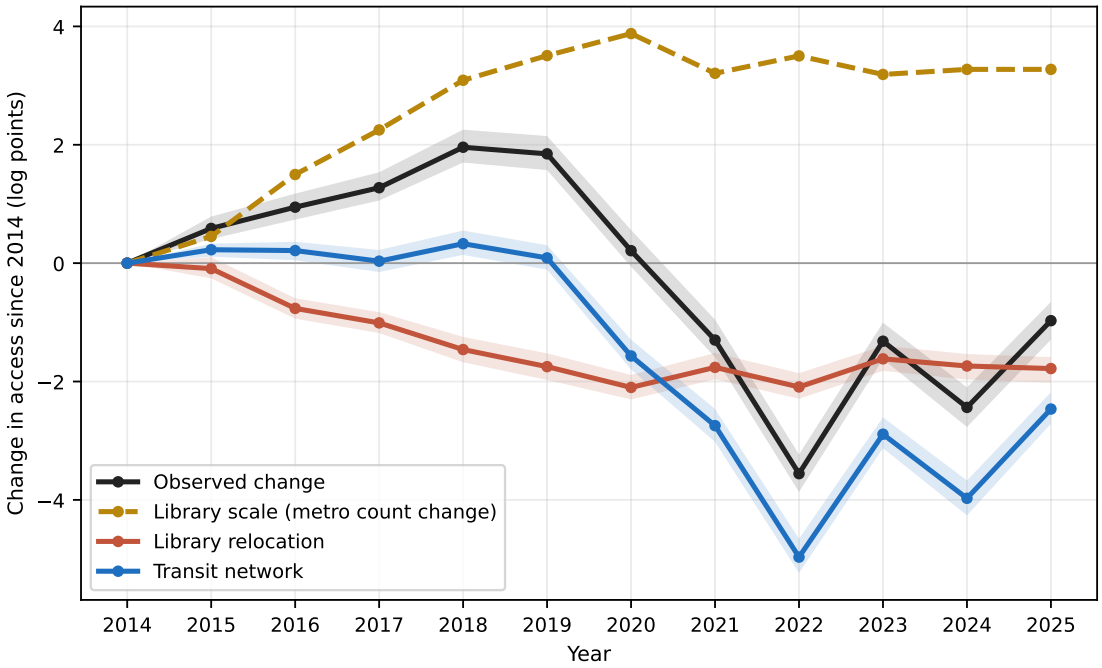
Full-service branches active in 2014 within the panel's block groups, quartiles of the location's 2014 45-minute transit count, and the share closed by 2025.

5.2 Libraries

Libraries are the network-driven service, the opposite of banks. In Figure 2 the observed line moves with the network line after 2019. The observed −1.0 splits into −2.5 network, +3.3 scale, and −1.8 relocation (all-metro companion: $-1.6 = -2.4 + 1.9 - 1.1$). The national count of library outlets barely changed (16,906 to 16,864 stationary outlets, 2014 to 2024), though the count in these metros rose. The network component exceeds the whole decline, while outlet growth (+3.3) more than offsets the adverse relocation (−1.8). The 2022 low (−5.0 log points) and the partial recovery are visible in the network component. An access measure without the decomposition would not show this, and attributing the decline to

library policy would be wrong.

Figure 2: Decomposition of 45-minute transit access to library outlets, 2014–2025.



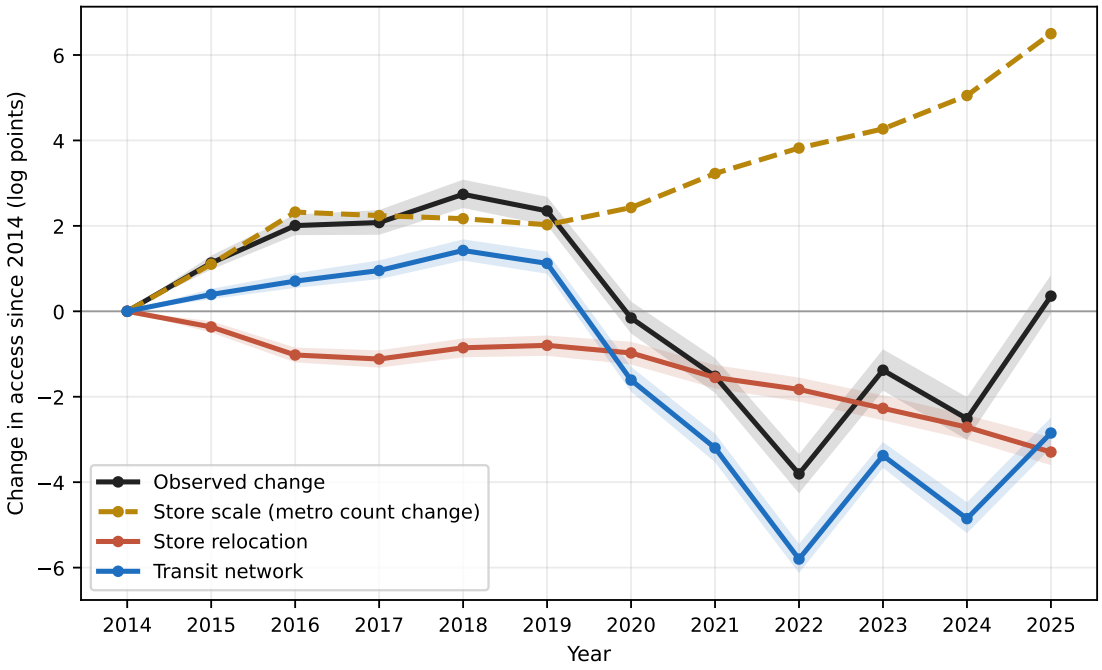
Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change.

In trajectory terms, transit access to stationary outlets fell **2.5%** [−2.8, −2.2] (flags included: −1.0%; with bookmobiles: identical; NYC included: −3.0%). The trajectory spans zero under metro clustering ([−5.1, +0.1]). The robust result for libraries is the network-led *mechanism*, not the magnitude.

5.3 Groceries

Groceries show a third mechanism. In Figure 3 the store-scale component grows (+6.5 in the clean-archive core; +8.4 in the all-metro companion) while the network declines and net store change favors transit-poor locations (a negative relocation term). The net change is small in both samples. The all-metro observed change is +1.9 ($= -2.7 + 8.4 - 3.8$), and in the clean-archive core it is +0.4 ($= -2.8 + 6.5 - 3.3$; 95% block-group bootstrap interval −0.1 to +0.8). The mechanism, not the size of the net change, is the robust fact.

Figure 3: Decomposition of 45-minute transit access to full-line grocery stores, 2014–2025.



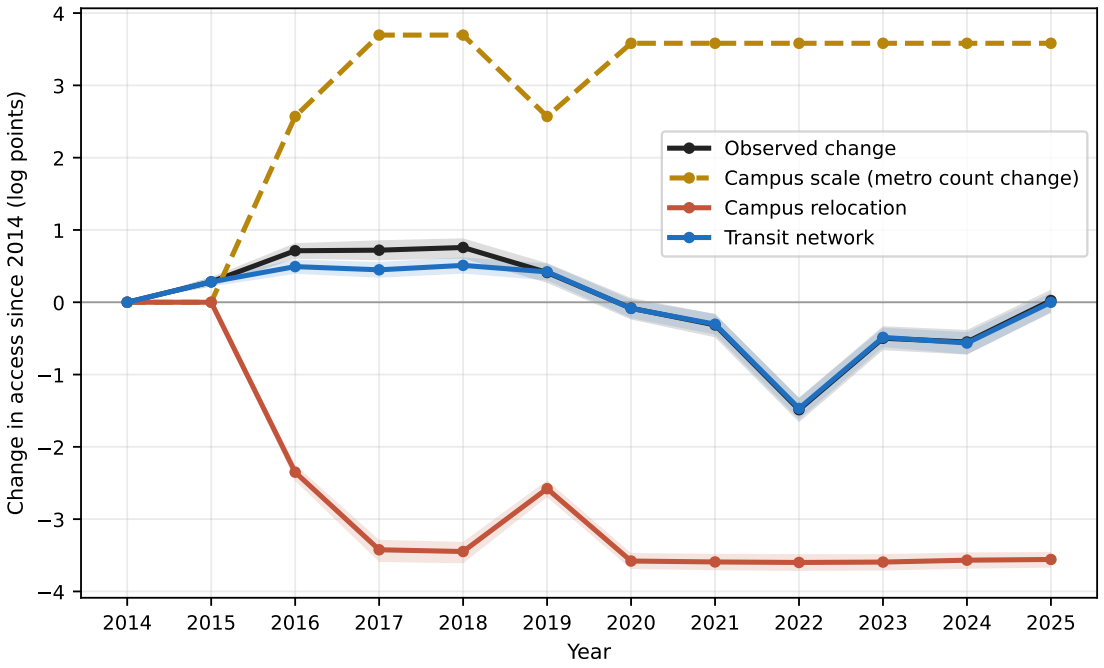
Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change.

In trajectory terms, on the panel, transit access to full-line grocery stores *rose* **1.7%** [+1.3, +2.2] (flags included: +3.8%; NYC included: +1.4%). Groceries are the only service whose panel-wide trajectory rises, though the metro-clustered interval spans zero ([−2.7, +6.4]). The all-SNAP variant is flat (−0.2%), so the gain is specific to full-line stores. It reflects composition within food retail, not growth of SNAP retail generally.

5.4 Community colleges

Community-college access shows near-zero net change, with modest offsetting destination components. In Figure 4 the observed 0.0 splits into 0.0 network, +3.6 campus scale, and −3.6 relocation (all-metro companion: $-0.3 = -0.3 + 1.8 - 1.8$). This section's numbers use the persistent-*campus* definition (Section 3). Counting records rather than campuses (the persistent-identity companion), the scale and relocation terms in the all-metro companion are −8.1 and +7.9, an artifact of multi-campus systems consolidating into single records (+2.8 and −2.7 in the clean-archive core).

Figure 4: Decomposition of 45-minute transit access to community-college campuses, 2014–2025.



Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change. Persistent-campus definition.

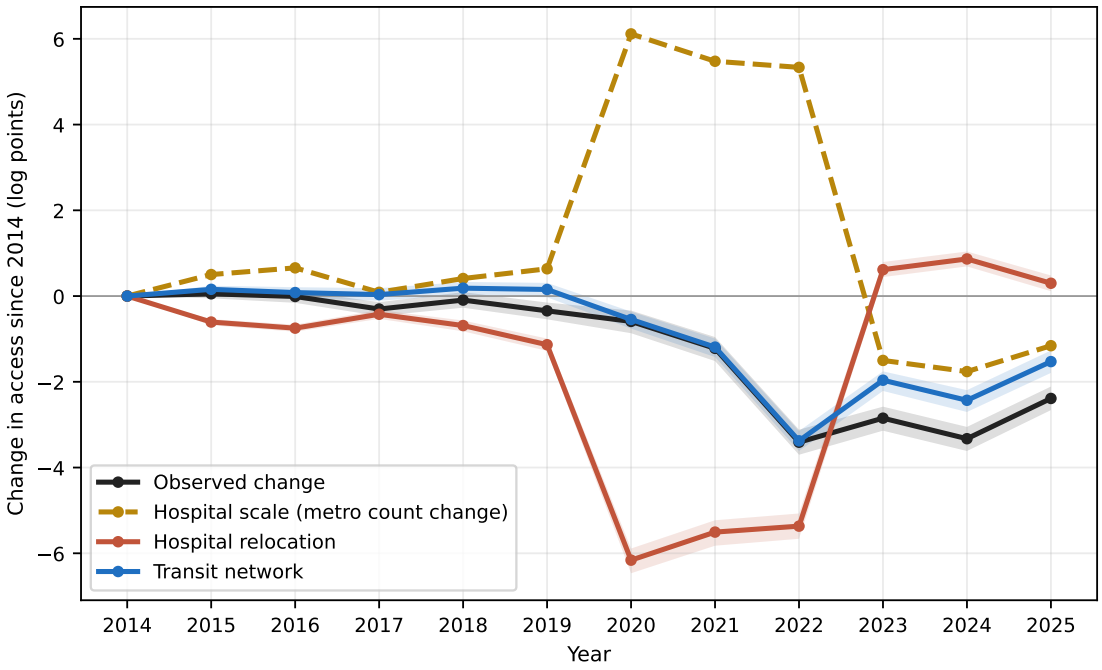
In trajectory terms, transit access to community-college campuses was nearly unchanged, at −0.4% [−0.6, −0.3] (flags included: +0.3%; NYC included: −0.5%; spanning zero under metro clustering, [−1.6, +0.7]). Under the same specification, definitions closer to the raw administrative categories produce larger declines. The persistent-identity companion (records, not campuses) gives −0.6%, and the annual-sector-code count gives −2.2%; neither additional decline reflects campus closures. Eleven metros with no community college in some panel year cannot be log-decomposed for this service (1.7% of the decomposition's block-group-quarters). Ten sit outside both named samples. The eleventh, Anchorage, is in the all-metro companion, so the community-college companion covers 34 metros and 55,099 block groups. Community colleges are also the sparsest service. In 87% of block-group-quarters no campus is reachable within 45 minutes, so level exhibits for this service rely on the coverage and nearest-time variants (Appendix Table A.5).

5.5 Hospitals

Hospital access changed through both sides, in proportions that depend on the sample. In the clean-archive core the network accounts for −1.5 of the observed −2.4, against a

net destination side of −0.9 (−1.2 hospital scale, +0.3 relocation; Figure 5). In the all-metro companion the destination side is somewhat larger than the network (−1.7 against −1.4; −3.1 = −1.4 − 4.8 + 3.0 within rounding). In both samples hospital counts fell, and a positive relocation term partly offset the fall. Hospitals are the one service whose destination components are small in the clean-archive core and several times larger in the all-metro companion. Both samples show a moderate total decline. The opposite-signed scale and relocation movements over 2020–22 in Figure 5 are the pandemic certification cohort entering and leaving the durable set. The observed line moves little because the two components offset.

Figure 5: Decomposition of 45-minute transit access to durable general acute hospitals, 2014–2025.



Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change.

In trajectory terms, transit access to durable general acute care hospitals (short-term acute plus critical access; the lead definition) fell **3.4%** [−3.6, −3.1] (flags included: −2.2%; NYC included: −3.8%), and the metro-clustered interval also excludes zero ([−5.2, −1.5]). Counting all durable hospitals, including psychiatric, rehabilitation, and long-term care, gives −3.7% [−4.0, −3.3]. The all-certification variant is indistinguishable at the endpoints and differs only in the 2020–22 path. Hospitals are sparse. The trajectory describes the roughly one quarter of residents who reach any durable hospital within 45 minutes, and the coverage and nearest-facility variants describe the rest (Appendix Table A.5).

6 Synthesis: mechanisms, incidence, and compounding disadvantage

6.1 One decomposition, six services

Table 3 assembles the endpoint decompositions on one scale (log points vs. the 2014 base; components sum exactly to observed in every row). The six rows, with jobs as the anchor, show six different balances of components: destination-led growth (jobs), destination-led decline partly offset by positive relocation (banks), network-led decline (libraries), destination growth against network decline and adverse relocation (groceries), near-zero change (community colleges), and a moderate decline split between both sides, with the balance depending on the sample (hospitals). The dispersion *within columns* is the paper's central fact. In the clean-archive core the network column runs from 0.0 to −6.4, while the destination columns range from −23.6 to +15.9. The all-metro companion, which has no jobs row, shows the same pattern for the five services (network −0.3 to −3.2; destination columns −19.9 to +8.4; Appendix Table A.8). Which side of access moves—the network or the destinations—is a property of the service, not of the era.

The same pattern holds when the panel is split at the pandemic. Over 2014–2019 the network component lies between −0.1 and +1.1 log points for every service, while the destination columns already differ by service (bank scale −8.3, grocery scale +2.0, jobs scale +10.5). Over 2019–2025 the network component is negative for every service (−0.4 to −6.3), while the destination columns again differ by service (bank scale −15.3, grocery scale +4.5). The network components of all services move together between the two periods, while the destination components differ by service in both.

Table 3: Cross-service endpoint decomposition.

Service	Observed	Network	Destination scale	Relocation
Jobs (anchor)	+15.2	−6.4	+15.9	+5.7
Bank branches	−21.4	−4.0	−23.6	+6.2
Public libraries	−1.0	−2.5	+3.3	−1.8
Groceries (full-line)	+0.4	−2.8	+6.5	−3.3
Community colleges	0.0	0.0	+3.6	−3.6
Durable general acute hospitals	−2.4	−1.5	−1.2	+0.3

2025 vs. 2014 annual average; clean-archive core; all entries in log points (100 times the change in $\ln(A+1)$), so the banks observed −21.4 corresponds to −19.3 percent. Components sum to observed; block-group bootstrap 95% CIs for observed, network, and relocation are within ±1.0 of the point estimates shown. Appendix A reports the all-metro companion for every service row. The jobs row is the companion paper's decomposition re-issued under the same sample rule (Arnold, 2026).

6.2 Who has access, and who lost it

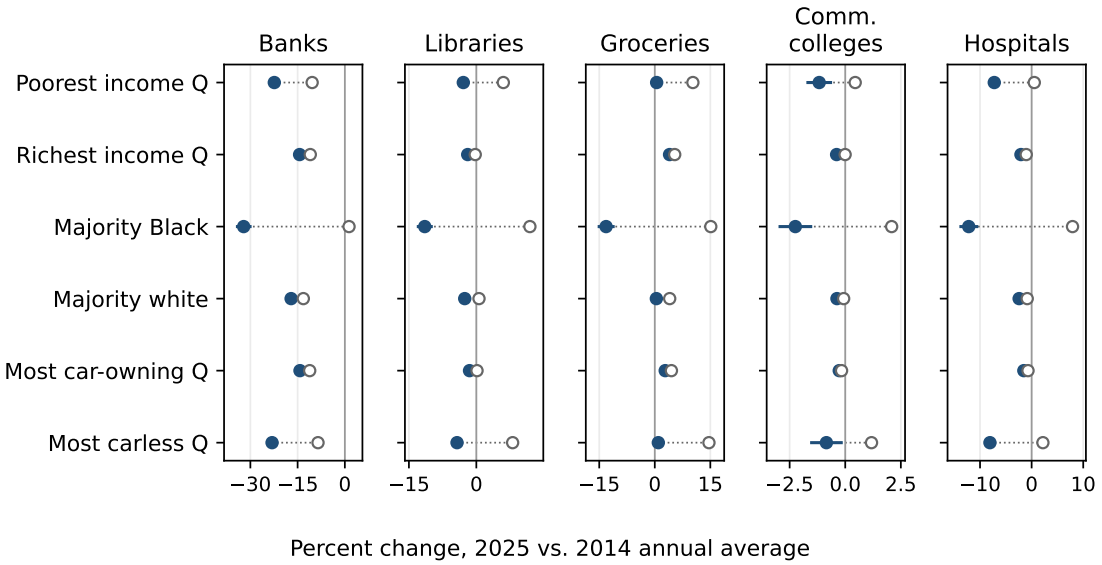
Demographic attributes throughout are the fixed 2018–22 five-year ACS vintage; level exhibits are quoted for 2023, the last year in which every layer, jobs included, is observed rather than held. Majority-race classes are population-share majorities; income quartiles are the inherited ACS block-group quartiles; carless quartiles are block-group quartiles of the carless-household share.

Levels. Outside New York, transit access to every essential service is higher in poor, majority-Black, and carless neighborhoods than in rich, white, car-owning ones. In 2023, richest-to-poorest income-quartile ratios run 0.35–0.53 across the six services, white-to-Black majority-neighborhood ratios run 0.35–0.56, and the most carless quartile of neighborhoods has five to nine times the access of the most car-owning quartile, depending on the service. The pattern reflects the concentration of transit service and destinations in central cities, and it means that the households most dependent on these services start with the highest access.

Change. Over 2014–2025 this advantage narrowed, and the incidence of the declines was regressive. Under the conservative (corrected) treatment, the largest access declines fall on majority-Black neighborhoods in every service (banks −32% against −17% in majority-white neighborhoods; libraries −11.5% against −2.6%; groceries −13.2% against +0.4%; hospitals −12.2% against −2.4%). They also fall on the poorest and most carless quartiles relative to the richest and most car-owning in nearly all services. Each quoted figure is the conservative side of the corrected/uncorrected bracket displayed in Figure 6. The characterizations replicate almost exactly under beginning-of-period (2009–13) classifications (U.S. Census Bureau, 2014), mapped from 2010 to 2020 block groups with the NHGIS crosswalk (Schroeder et al., 2026) (majority-Black bank change −32.3% against majority-white −17.3%; Appendix Table A.3), so the incidence does not depend on who moved in after access changed.

Measurement. For advantaged groups the corrected and uncorrected estimates differ by at most 3.9 points (majority-white banks). For disadvantaged groups they diverge by up to 33 points (majority-Black banks: −32 to +1). Archive coverage changes concentrate in the agencies serving disadvantaged areas, so the group *differences* are sensitive to the archive-entry correction treatment in a way the aggregate results are not. The width of the majority-Black bank bracket is traceable to a single metro's archive entry, and the corrected series is the one stable to it (Appendix Table A.3). We therefore quote group-level changes as bounds, read the conservative side as the incidence estimate, and treat the asymmetry as a finding in its own right. The neighborhoods whose access history is hardest to measure are the disadvantaged ones.

Figure 6: Access change by neighborhood group and service, corrected and uncorrected.



Percent change in the 45-minute count, 2025 vs. the 2014 base; the panel (52 metros ex-NYC), population-weighted, flagged cells excluded; filled = corrected with 95% block-group-clustered CIs, open = uncorrected upper bound; panel x-scales differ by service.

6.3 Compounding disadvantage

Do the same neighborhoods see access decline in several services at once? We call a neighborhood a loser of service s if its within-neighborhood corrected access change from the 2014 base to 2025 is -5 log points or worse. We then compare multi-service loss rates to the benchmark in which each service's losses (at their observed marginal rates) strike independently. The exercise runs on the clean-archive core. In the all-metro companion the multipliers for four or more and for five or more services are $\times 5.3$ and $\times 28.6$, within 10 percent of the core values in Table 4. Shares and multipliers count block groups unweighted. Loss marginals differ widely by service (banks 55%, jobs 31%, groceries 28%, libraries 17%, hospitals 11%, community colleges 3%), which the benchmark respects.

Table 4 reports the result. Broad co-loss (two or more services) occurs below the independence rate ($\times 0.84$), and the multiplier rises from $\times 1.6$ at three services to $\times 31.5$ at five or more. The rise survives four re-benchmarks, each aimed at a different artifact: (i) against each metro's own loss marginals, which absorbs between-metro composition, $\times 3.5$ at four or more services and $\times 13$ at five or more; (ii) under population weighting, $\times 6.9$ and $\times 44$; (iii) dropping community colleges (the sparsest layer) or jobs entirely, $\times 6.2$ and $\times 41$ without community colleges and $\times 13$ and $\times 57$ without jobs (at four or more and five or more of the remaining five services); (iv) dropping the three tail metros (Seattle, Chicago, Denver), $\times 9.6$ and $\times 73$, so the pattern does not depend on those three metros. It does not

survive holding the network component fixed. Within network-loss strata, which absorb the network component every service shares in a neighborhood, the multipliers fall to ×1.08 at four or more services (95% interval 0.99 to 1.14) and ×1.33 at five or more (1.17 to 1.42). Jointly within metro-by-stratum cells they fall to ×1.01 and ×1.19. Deep co-loss is therefore mostly the network component that the services share. Where the transit network declined, access to several services fell at once. The small excess that remains at five or more services can reflect sub-metro clustering of destination losses, as when several services close in the same declining commercial district. Metro-clustered intervals are percentile bootstraps over metros (500 draws).

Table 4: Multi-service loss against the independence benchmark.

Services lost	Observed	Indep.	Mult.	Metro 95% CI	Within strata	Within metro
≥ 2 of 6	38.6%	45.7%	×0.84	—	—	—
≥ 3 of 6	22.3%	13.9%	×1.6	—	—	—
≥ 4 of 6	11.9%	2.15%	×5.5	[3.7, 9.8]	×1.08	×3.5
≥ 5 of 6	4.63%	0.147%	×31.5	[16.2, 76.4]	×1.33	×13.0
All six	0.455%	0.0029%	×157	[65.0, 580.9]	×1.47	×40.4

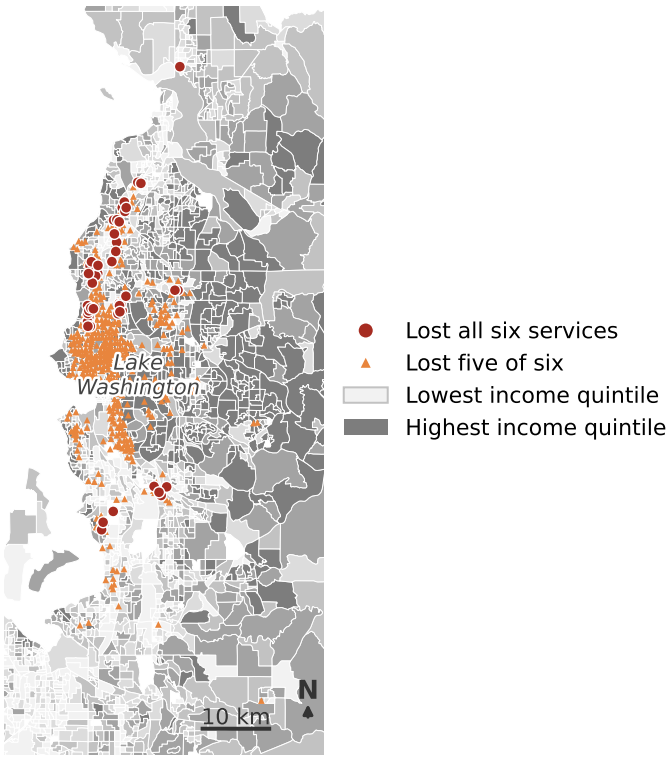
Clean-archive core. Losses: corrected 2014–2025 change ≤ -5 log points. Strata = the block groups whose mean network component across the six services is exactly zero (36%), and quartiles of the rest; within-metro = each metro's own loss marginals. Conditional multipliers pool observed over expected losers across strata or metros, each expected count from that group's own loss marginals. Metro-clustered intervals and conditional multipliers are computed for the deep thresholds (≥ 4) only.

The extreme tail shows the same mechanism in its strongest form. The 128 neighborhoods (0.45%) that lost *all six* services are ×157 over independence unconditionally and ×1.47 within network strata, and 72% of them are in three metros (Seattle, Chicago, Denver). In the all-metro companion the tail is concentrated in Boston, Seattle, and Philadelphia, and the multiplier falls from ×171 to ×1.75 within strata. Losing several services at once therefore mostly indicates localized network collapse, a network change that removes access to many services together, rather than destinations closing independently in the same places. The network reading is consistent with agency-reported service. Across the 19 core metros, the banks network component correlates at $r = 0.50$ with the change in National Transit Database (NTD) vehicle revenue miles (Federal Transit Administration, 2026), both sides restricted to the same metros (Appendix Table A.3). The check is not fully independent of the sample, because disagreement with these revenue miles is one of the criteria that flag archive quarters (Appendix Table A.6). In Seattle, revenue miles rose while measured 45-minute reach fell steeply, which indicates a restructuring of connectivity rather than a reduction in total service. The agency record matches, with bus routes replaced, revised, or reoriented to Link stations at the 2021 Northgate and 2024 Lynnwood openings (King

County Metro, 2021, 2024). In Denver, revenue miles and measured reach moved together. The deep-tail multipliers are larger under a stricter −10 log-point definition, and the pattern also appears in absolute-threshold form. Entering three or more of the bank, library, grocery, and hospital "deserts" simultaneously (from positive 2014 access to zero in 2025) is 8.3 times the independence benchmark, though it affects only 0.10% of neighborhoods.

Figure 7 maps the deep tail in Seattle, the metro with the largest share of the all-six tail. The deep losers are concentrated in King County (449 of its 1,542 block groups lost five or more services), with 51 in Snohomish County and none in Pierce County. Within King County, 41.5% of block groups west of Lake Washington (west of 122.20°W, the lake's eastern shore) lost five or more services, against 7.3% east of it. The concentration matches the network mechanism of the all-six tail. Block groups losing five or more services have three times the carless share of those losing none (12.5% against 4.1%) but not lower incomes. Only the 48 that lost all six skew to the lowest income quintiles.

Figure 7: Neighborhoods losing five or six services, Seattle metro.



Income quintile shading (2018–22 ACS; darker = higher income); orange = lost five of six services (452 block groups); dark red = lost all six (48); losses = corrected 2014–2025 change ≤ −5 log points; all 2,733 every-year block groups covered (the metro routes 2,734; one lacks every-year status).

Who are the multi-losers? From neighborhoods losing no service to those losing all six, the low-income population share rises from .21 to .29, the carless household share from

.04 to .12, the Black population share from .06 to .13, and majority-minority status from 34% to 44%. Median household income falls from $106,000 to $89,000, and distance to the metro core falls from 31 to 18 km. The carless share and distance to the core change monotonically through four services lost, and the low-income share, Black share, and median income through three. Beyond those points the gradients level off or partly reverse; counts are small at five and six services. The pattern replicates under beginning-of-period (2009–13) classifications (low-income share .25 to .39, carless .04 to .13, median income $75,000 to $56,000). Deep multi-service loss is concentrated in poor, carless, minority, relatively central neighborhoods, where transit dependence is highest. *Broad* co-loss is below independence; *deep* co-loss (four or more services) is far above it.

7 New York: the exception

New York is excluded from the paper's samples and reported here instead. All NYC decompositions are uncorrected and stated as such. Four facts justify the treatment.

Scale. New York contributes 14,472 every-year block groups (observed and unflagged in every year; 19.8 million residents), half the size of the clean-archive core (28,147 block groups; 41.8 million residents). Pooled person-weighted results would give New York about a third of the weight.

Grocery and hospital access change more. Because the archive-entry correction is unidentified for a single metro, the only like-for-like comparison is between New York's uncorrected changes and the clean-archive core's uncorrected changes (Table 5). Groceries and hospitals differ most. New York's grocery access rose +10.1 log points against +0.7 in the core, and its hospital access fell −11.7 against −2.3. Libraries fell −2.6 against −0.5, and community colleges −0.8 against 0.0. Banks are the exception, with similar changes in New York (−19.7) and the core (−20.9).

Table 5: New York against the clean-archive core, like-for-like.

Service	New York	Clean-archive core
Bank branches	−19.7	−20.9
Public libraries	−2.6	−0.5
Groceries (full-line)	+10.1	+0.7
Community colleges	−0.8	0.0
Durable general acute hospitals	−11.7	−2.3

Uncorrected endpoint change in the 45-minute count, 2025 vs. the 2014 annual average, log points; New York = 14,472 every-year block groups; population-weighted.

The decompositions are more destination-driven. The bank decomposition shows the

core's pattern in stronger form. The network contributes essentially nothing (+0.7), the branch decline is steeper (−28.3 against −23.6), and the relocation term is larger (+7.9 against +6.2). The grocery gain is nearly all store growth (+12.1 scale against −2.0 relocation), and the library decline comes from location rather than the network (−2.0 relocation, network ≈ 0). In New York the routed network barely changed, so most of the access change comes from the destination side.

The race gradient reverses for some services. Outside New York, access *levels* favor poor and minority neighborhoods for every service (Section 6.2). In New York the income and race gradients are flatter, and the race gradient reverses sign for jobs, banks, and community colleges (white/Black 1.08–1.26; the jobs and banks intervals include parity). Majority-white neighborhoods there reach more jobs, banks, and community colleges by transit, the opposite of the pattern outside New York. Any pooled "who has access" exhibit would average opposite-signed race gradients for those services.

8 Conclusion

Between 2014 and 2025, transit access to essential services in the 52 metros outside New York changed in different directions and for different reasons. The trajectories cover all 52 metros; the decompositions and the compounding analysis cover the clean-archive core of 19 metros. Bank access fell 18.1%, mainly because branches closed. Library access fell 2.5%, because the transit network changed. Hospital access fell 3.4% through both the network and the destinations. Grocery access rose 1.7% on the panel, as store growth offset network decline and a shift of stores toward transit-poor locations. Community-college access was nearly unchanged once reclassified and consolidated colleges are not counted as closures. Only the bank and hospital declines have metro-clustered intervals that exclude zero, and the smaller changes describe these metros rather than a national process. Losses also compound, mostly through the transit network the services share. Neighborhoods losing four or more of the six services, jobs included, are 5.5 times as common as independent losses would imply, and those losing five or more are 31.5 times as common. New York is reported separately in Section 7.

The decomposition separates two kinds of change. Before 2020 the network component was close to zero for every service, between −0.1 and +1.1 log points, while the destination components already differed by service. After 2019 the network component was negative for every service, from −0.4 to −6.3 log points, and the destination components still differed by service. The period moved every service's network component in the same direction, while the destination side followed each service's own industry: branch consolidation, supermarket

growth, and hospital openings and closures. Library and bank access both fell, but the library decline came from the network and the bank decline from branch closures, so the two call for different responses. The distribution of the changes is a second result. Under the archive-entry correction, majority-Black neighborhoods, which started with more access than majority-white ones, had the largest declines in every essential service. The poorest and most carless neighborhoods had larger declines than the richest and most car-owning in nearly all services. These are also the neighborhoods whose access history is hardest to measure, because archive coverage changed most for the agencies that serve them, above all Philadelphia's SEPTA.

The results have three implications for practice. First, the decomposition identifies which institution can act. Where the network component dominates, as for libraries and in the network-collapse corridors, the relevant policy is transit service. Where the destination side dominates, as for banks, transit policy cannot reverse the trend, but siting policy can affect it. The positive bank relocation term reflects redundancy. Neighborhoods that reached many branches absorbed closures, while a single closure could leave a neighborhood that reached few with none. Siting and community-reinvestment review could target the closures that leave a neighborhood with no reachable branch. Second, access to essential services can be monitored with public data. The inputs are agencies' published schedules and federal facility files, and the travel-time matrices that measure job access serve every service without additional routing. Third, the neighborhoods losing many essentials at once are identifiable and concentrated where transit dependence is highest. In King County, 41.5% of block groups west of Lake Washington lost five or more services, against 7.3% east of it, and Figure 7 shows where those neighborhoods are.

The measure has limits. It is potential access and does not account for facility capacity, service quality, or frequency and transfer burden beyond their effect on median travel time. It is measured at the morning peak (Section 4) on an archive that favors larger agencies (Section 3.3), and destination vintages are annual, so within-year destination movement is not observed. At one of the twelve archive-entry events, Santa Rosa in 2017, agency-reported service rose by 5 log points or more along with archive coverage, so there the archive-entry correction may remove some real gain.

The analysis is descriptive. It records how access changed and which side moved, not whether the changes affected banking, food security, health, or education. For studies of those outcomes, the panel provides a neighborhood-level measure of access to five services that is comparable over time.

Over 2014–2025, the side of access that moved depended on the service. Tracking only transit service would have missed the branch closures. Tracking only facility counts would

have missed the network losses that removed several services at once from the neighborhoods that depend most on transit.

References

Robert M. Adams and Shane M. Sherlund. Where's the bank? banking access in the era of branch consolidation. Technical Report 2025-086, Federal Reserve Board, 2025.

Jeff Allen and Steven Farber. Planning transport for social inclusion: An accessibility-activity participation approach. *Transportation Research Part D: Transport and Environment*, 78:102212, 2020. doi: 10.1016/j.trd.2019.102212.

Sierra Arnold. Public transportation access and food insecurity. *Journal of Regional Science*, 65(5):1449–1467, 2025. doi: 10.1111/jors.70010.

Sierra Arnold. The evolution of transit access to jobs in american cities. Working paper, Xavier University, 2026.

Geneviève Boisjoly and Ahmed El-Geneidy. Daily fluctuations in transit and job availability: A comparative assessment of time-sensitive accessibility measures. *Journal of Transport Geography*, 52:73–81, 2016. doi: 10.1016/j.jtrangeo.2016.03.004.

R. Blake Buchalter, Paul R. Gunsalus, Madeleine M. Blazel, Michael W. Kenyhercz, and Jarrod E. Dalton. Disparities in multi-modal spatial access to primary and specialty care in U.S. neighborhoods: A cross-sectional and temporal analysis of health services. *PLoS One*, 20(9):e0330427, 2025. doi: 10.1371/journal.pone.0330427.

Centers for Medicare and Medicaid Services. Provider of Services File (Hospital and Non-Hospital Facilities), December Vintages 2013–2025 [dataset]. data.cms.gov, Office of Enterprise Data and Analytics, 2026. URL https://data.cms.gov/provider-characteristics/hospitals-and-other-facilities/provider-of-services-file-quality-improvement-and-evaluation-system. Now titled "Provider of Services File – Quality Improvement and Evaluation System." Accessed July 10, 2026.

M. W. Conway, A. Byrd, and M. van Eggermond. Accounting for uncertainty and variation in accessibility metrics for public transport sketch planning. *Journal of Transport and Land Use*, 11(1):541–558, 2018. doi: 10.5198/jtlu.2018.1074.

Mengying Cui and David Levinson. Primal and dual access. *Geographical Analysis*, 52(3): 452–474, 2020. doi: 10.1111/gean.12220.

Francis P. Donnelly. Regional variations in average distance to public libraries in the United States. *Library & Information Science Research*, 37(4):280–289, 2015. doi: 10.1016/j.lisr.2015.11.008.

Alireza Ermagun and Frank Witlox. Transit access effectiveness in American metropolitan areas. *Journal of Transport Geography*, 116:103853, 2024. doi: 10.1016/j.jtrangeo.2024.103853.

Alireza Ermagun, Fatemeh Janatabadi, and Frank Witlox. Beyond the 15-minute city dichotomy: Time-denominated access to essential services in Chicago. *Transportation Research Part A: Policy and Practice*, 195:104428, 2025. doi: 10.1016/j.tra.2025.104428.

Steven Farber, Melinda Z. Morang, and Michael J. Widener. Temporal variability in transit-based accessibility to supermarkets. *Applied Geography*, 53:149–159, 2014. doi: 10.1016/j.apgeog.2014.06.012.

Federal Deposit Insurance Corporation. Summary of Deposits, 2013–2025 [dataset]. FDIC BankFind Suite, 2025. URL https://banks.data.fdic.gov/bankfind-suite/SOD. Annual survey of branch office deposits as of June 30. Accessed July 8, 2026.

Federal Transit Administration. National Transit Database, Complete Monthly Ridership (with Adjustments and Estimates) [dataset]. U.S. Department of Transportation open data portal, dataset 8bui-9xvu, 2026. URL https://data.transportation.gov/d/8bui-9xvu. Monthly vehicle revenue miles by agency, mode, and type of service. Accessed July 6, 2026.

Nicole Foth, Kevin Manaugh, and Ahmed El-Geneidy. Towards equitable transit: Examining transit accessibility and social need in Toronto, Canada, 1996–2006. *Journal of Transport Geography*, 29:1–10, 2013. doi: 10.1016/j.jtrangeo.2012.12.008.

Karst T. Geurs and Bert van Wee. Accessibility evaluation of land-use and transport strategies: Review and research directions. *Journal of Transport Geography*, 12(2):127–140, 2004. doi: 10.1016/j.jtrangeo.2003.10.005.

Joe Grengs. Nonwork accessibility as a social equity indicator. *International Journal of Sustainable Transportation*, 9(1):1–14, 2015. doi: 10.1080/15568318.2012.719582.

Susan L. Handy and Debbie A. Niemeier. Measuring accessibility: An exploration of issues and alternatives. *Environment and Planning A*, 29(7):1175–1194, 1997. doi: 10.1068/a291175.

Walter G. Hansen. How accessibility shapes land use. *Journal of the American Institute of Planners*, 25(2):73–76, 1959. doi: 10.1080/01944365908978307.

Nicholas W. Hillman. Geography of college opportunity: The case of education deserts. *American Educational Research Journal*, 53(4):987–1021, 2016. doi: 10.3102/0002831216653204.

Homeland Infrastructure Foundation-Level Data. Hospitals [dataset]. HIFLD Open hospitals layer, copy hosted on ArcGIS Online, 2020. URL https://www.arcgis.com/home/item.html?id=fbd736551b834a069258dda168b7c95f. 7,581 facilities; source dates through August 13, 2019. Accessed July 13, 2026.

Institute of Museum and Library Services. Public Libraries Survey, Public Library Outlet Data Files, Fiscal Years 2013–2024 [dataset]. Institute of Museum and Library Services, Washington, DC, 2026. URL https://www.imls.gov/research-evaluation/surveys/public-libraries-survey-pls. Accessed July 9, 2026.

Fatemeh Janatabadi and Alireza Ermagun. Unraveling transit service and land use components of the socio-spatial inequality of access. *Environment and Planning B: Urban Analytics and City Science*, 51(7):1447–1462, 2024a. doi: 10.1177/23998083231207534.

Fatemeh Janatabadi and Alireza Ermagun. Four-step framework for untangling place and mobility components of access: An equal opportunity perspective. *Journal of Transport Geography*, 116:103831, 2024b. doi: 10.1016/j.jtrangeo.2024.103831.

Javad Jomehpour Chahar Aman and Janille Smith-Colin. Transit deserts: Equity analysis of public transit accessibility. *Journal of Transport Geography*, 89:102869, 2020. doi: 10.1016/j.jtrangeo.2020.102869.

Armita Kar, Andre L. Carrel, Harvey J. Miller, and Huyen T. K. Le. Public transit cuts during COVID-19 compound social vulnerability in 22 US cities. *Transportation Research Part D: Transport and Environment*, 110:103435, 2022. doi: 10.1016/j.trd.2022.103435.

Alex Karner. Assessing public transit service equity using route-level accessibility measures and public data. *Journal of Transport Geography*, 67:24–32, 2018. doi: 10.1016/j.jtrangeo.2018.01.005.

King County Metro. October 2021 Service Change: North Link Connections Mobility Project. Presentation to the Seattle Transit Advisory Board, August 26, 2021, 2021. URL https://www.seattle.gov/documents/departments/sdot/transitprogram/tab/northlink_tab_presentation082621.pdf. Accessed November 3, 2026.

King County Metro. Lynnwood Link Connections. Mobility project documentation, King County, Washington, 2024. URL https://kingcounty.gov/en/dept/metro/programs-and-projects/lynnwood-link-connections. Accessed 2026-11-03.

Willem Klumpenhouwer, Jeff Allen, Lisa Li, Rick Liu, Mitchell Robinson, Diego Da Silva, Steven Farber, Alex Karner, Dana Rowangould, Amer Shalaby, Mary Buchanan, and Steven Higashide. A comprehensive transit accessibility and equity dashboard. *Findings*, July 2021. doi: 10.32866/001c.25224.

Alexander Li, Aurimas Racas, Junaid Syed, Mengyang Liu, Przemyslaw Zientala, and Tejas Santanam. Ease and equity of point of interest accessibility via public transit in the U.S. arXiv:2212.06954, 2022. Revised 2025.

Karen Lucas. Transport and social exclusion: Where are we now? *Transport Policy*, 20: 105–113, 2012. doi: 10.1016/j.tranpol.2012.01.013.

Wei Luo and Fahui Wang. Measures of spatial accessibility to health care in a GIS environment: Synthesis and a case study in the Chicago region. *Environment and Planning B: Planning and Design*, 30(6):865–884, 2003. doi: 10.1068/b29120.

Karel Martens. Justice in transport as justice in accessibility: Applying Walzer's 'Spheres of Justice' to the transport sector. *Transportation*, 39(6):1035–1053, 2012. doi: 10.1007/s11116-012-9388-7.

National Center for Education Statistics. Integrated Postsecondary Education Data System, Institutional Characteristics: Directory Information Files, 2013–2024 [dataset]. IPEDS Data Center, U.S. Department of Education, 2025. URL https://nces.ed.gov/ipeds/datacenter/DataFiles.aspx. Files HD2013–HD2024. Accessed July 9, 2026.

Tijs Neutens. Accessibility, equity and health care: Review and research directions for transport geographers. *Journal of Transport Geography*, 43:14–27, 2015. doi: 10.1016/j.jtrangeo.2014.12.006.

Andrew Owen, Shirley Shiqin Liu, and Eric M. Lind. Access across america: Transit 2024. Technical Report CTS 25-18, University of Minnesota Accessibility Observatory, 2025. URL https://hdl.handle.net/11299/277743.

Rafael H. M. Pereira, Tim Schwanen, and David Banister. Distributive justice and equity in transportation. *Transport Reviews*, 37(2):170–191, 2017. doi: 10.1080/01441647.2016.1257660.

Pew Research Center. 1 in 10 Americans rarely or never drive a car, 2024. URL https://www.pewresearch.org/short-reads/2024/11/14/1-in-10-americans-rarely-or-never-drive-a-car/. Short Read, November 14 (nationally representative survey of 5,410 U.S. adults, August 12–18, 2024).

Piotr Rosik, Marcin Mazur, Marcin Stępniak, Tomasz Komornicki, Sławomir Goliszek, Patryk Duma, and Paweł Churski. What drives potential accessibility decomposition? *Journal of Transport and Land Use*, 18(1):1–27, 2025. doi: 10.5198/jtlu.2025.2508.

Manuel Santana Palacios and Ahmed El-Geneidy. Cumulative versus gravity-based accessibility measures: Which one to use? *Findings*, February 2022. doi: 10.32866/001c.32444.

Jonathan Schroeder, David Van Riper, Steven Manson, Grace Cooper, Zachary Krause, Tracy Kugler, Tsu Zhu, and Steven Ruggles. IPUMS National Historical Geographic Information System: Version 21.0 [dataset]. Minneapolis, MN: IPUMS, 2026.

Anthony F. Shorrocks. Decomposition procedures for distributional analysis: A unified framework based on the Shapley value. *Journal of Economic Inequality*, 11(1):99–126, 2013. doi: 10.1007/s10888-011-9214-z.

Marcin Stępniak and Piotr Rosik. The role of transport and population components in change in accessibility: The influence of the distance decay parameter. *Networks and Spatial Economics*, 18(2):291–312, 2018. doi: 10.1007/s11067-017-9376-8.

U.S. Census Bureau. American Community Survey, 5-Year Estimates, 2009–2013: Detailed Tables [dataset]. Census Data API, 2014. URL https://api.census.gov/data/2013/acs/acs5. Released December 4, 2014.

U.S. Census Bureau. American Community Survey, 5-Year Estimates, 2018–2022: Detailed Tables [dataset]. Census Data API, 2023a. URL https://api.census.gov/data/2022/acs/acs5. Released December 7, 2023.

U.S. Census Bureau. American Community Survey, 1-Year Estimates [dataset], 2023b. Table B25044, Tenure by Vehicles Available.

U.S. Census Bureau. County Business Patterns: 2023 [dataset]. U.S. Census Bureau, CBP Datasets, 2025. URL https://www.census.gov/data/datasets/2023/econ/cbp/2023-cbp.html. Released June 26, 2025.

U.S. Census Bureau. LEHD Origin-Destination Employment Statistics, Version 8: Workplace Area Characteristics Files, 2014–2023 [dataset], 2026. URL https://lehd.ces.census.gov/data/lodes. Accessed June 13, 2026.

U.S. Department of Agriculture, Food and Nutrition Administration. SNAP Retailer Locator Historical Data, 2005–2025 [dataset]. USDA Food and Nutrition Administration, 2026. URL https://www.fna.usda.gov/snap/retailer-locator/data. Data current as of December 31, 2025. Accessed July 9, 2026.

Michele Ver Ploeg, Vince Breneman, Tracey Farrigan, Karen Hamrick, David Hopkins, Phillip Kaufman, Biing-Hwan Lin, Mark Nord, Travis A. Smith, Ryan Williams, Kelly Kinnison, Carol Olander, Anita Singh, and Elizabeth Tuckermanty. Access to affordable and nutritious food: Measuring and understanding food deserts and their consequences. Report to Congress, Administrative Publication 036, USDA Economic Research Service, 2009.

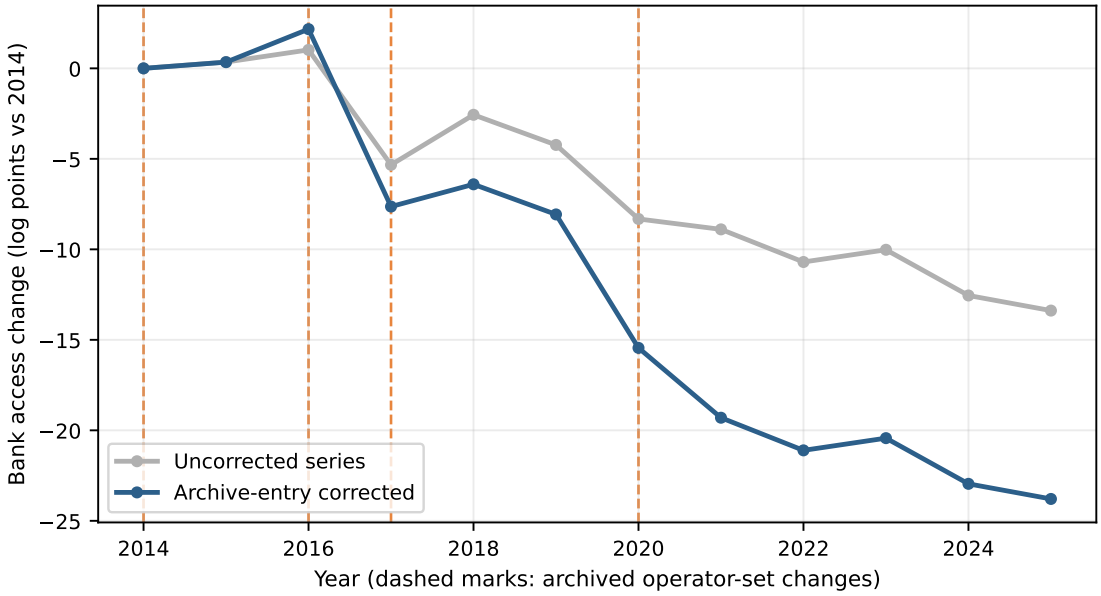
Martin Wachs and T. Gordon Kumagai. Physical accessibility as a social indicator. *Socio-Economic Planning Sciences*, 7(5):437–456, 1973. doi: 10.1016/0038-0121(73)90041-4.

Michael J. Widener, Steven Farber, Tijs Neutens, and Mark Horner. Spatiotemporal accessibility to supermarkets using public transit: An interaction potential approach in Cincinnati, Ohio. *Journal of Transport Geography*, 42:72–83, 2015. doi: 10.1016/j.jtrangeo.2014.11.004.

Y. Zhang, F. Xu, E. Talen, M. C. González, and Y. Li. Quantifying racial inequality in transit access across New York City. *PNAS Nexus*, 5(5):pgag025, 2026. doi: 10.1093/pnasnexus/pgag025.

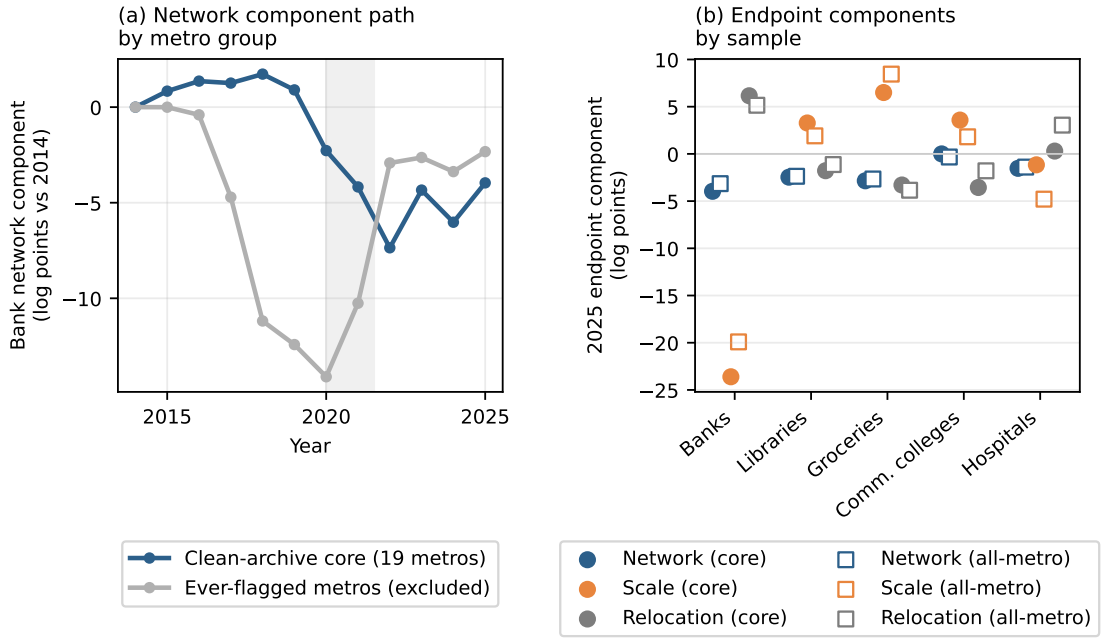
Appendix A Supporting exhibits

Figure A.1: The archive-entry correction in one metro (Cincinnati, bank access).



Uncorrected and corrected series, log points vs. 2014; vertical marks = archived-operator-set changes; population-weighted every-year block groups. An archive regime is a maximal span of quarters over which a metro's set of archived feeds is stable; boundaries are dated from the companion panel's feed inventory (Arnold, 2026), and a metro has two to eleven regimes. For each series, each block group's cumulative cross-regime jump against its 2014 base is removed from the observed and network sides equally, preserving exact additivity. Corrections are nonzero for 2–27% of block-group years, depending on the service. Validation against agency-reported revenue miles is the archive-entry row of Table A.3.

Figure A.2: Component paths by sample.



(a) Banks network component, clean-archive core vs. excluded metros; (b) 2025 endpoint components, clean-archive core vs. all-metro companion; population-weighted, corrected, log points vs. 2014. Sample definitions. *The panel*: the balanced set of 52 metros outside New York with at least 36 of 48 quarters routed; all trajectory and group-level estimates. *The clean-archive core*: the panel's every-year block groups (those observed and unflagged in every year) in the 19 never-flagged metros observed in every year from 2014 to 2025, 28,147 block groups, constant by construction; the same rule defines the companion paper's re-issued jobs decomposition. *The all-metro companion*: the panel's every-year block groups in every metro that has them, which is 35 of the 52 metros (the 19 core metros and 16 ever-flagged metros; 55,359 block groups). The core holds 32.5% of the panel's population and skews toward mid-sized systems; its roster is Table A.1, and Table A.8 reports both samples' decompositions. Metros are excluded at the metro-era level by the quality flags of Table A.6.

Table A.1: Panel metros: summary.

	Metros
Routed	96
In the analysis panel	92
With merged census attribute values	90
Balanced panel (ex-NYC; ≥36 of 48 quarters)	52
Clean-archive core (never flagged, observed 2014–2025)	19

The 92-metro roster of the analysis panel (quarters covered, block groups, population) is Table S1 of the online supplementary material. The clean-archive core comprises Albuquerque, Austin, Champaign–Urbana, Charleston, Chicago, Cincinnati, Denver, Kansas City, Madison, Oxnard, Phoenix, Portland (OR), Rochester, Sacramento, San Antonio, Santa Rosa, Seattle, Spokane, and Stockton. Four routed metros (Fairbanks, Jacksonville, Palm Bay, Riverside–San Bernardino) have no car-travel matrices and are absent from the analysis panel; the two Connecticut metros are in the analysis panel, but their attribute values did not merge.

Table A.2: Hospital layer: active facilities and geocode coverage by year.

Year	Active hospitals	Geocoded	Coverage
2013	7,193	6,565	91.3%
2014	7,181	6,551	91.2%
2015	7,168	6,542	91.3%
2016	7,175	6,552	91.3%
2017	7,164	6,548	91.4%
2018	7,120	6,508	91.4%
2019	7,063	6,448	91.3%
2020	7,241	6,602	91.2%
2021	7,298	6,651	91.1%
2022	7,270	6,629	91.2%
2023	7,044	6,421	91.2%
2024	7,038	6,411	91.1%
2025	7,023	6,391	91.0%
Geocode sources (unique addresses): Census 7,684, HIFLD 789, OSM 251.			

December Provider of Services vintages, category-01 facilities. Three universes nest: 8,440 distinct certifications ever active as category-01 hospitals across the vintages; roughly 7,000–7,300 active in any December vintage (this table); and the durable general acute subset of Table 1 that defines the lead. Geocoding is a four-source cascade: Census batch and single-address geocoders, the HIFLD facility layer, and OpenStreetMap. The ungeocoded set is persistent, not churning. Of all certifications, 91.2% geocode in every year they appear, and only 1.1% ever switch status.

Table A.3: External checks on the destination layers, the network component, and the demographic attributes.

Check	Comparison	Result
Groceries vs. Census CBP 445110	state store counts, 2023	corr. 0.976 (logs); median ratio 0.78
Hospitals vs. HIFLD	state durable-GA counts	corr. 0.997 (logs, 56 units); median ratio 0.90
Libraries vs. PLS system files	outlet counts	exact (1.000)
Network component vs. NTD	Δlog revenue miles, 19 core metros	corr. 0.50
Archive-entry correction vs. NTD	Δlog VRM over the 12 months either side of 12 archive-entry events (regime changes moving access $\geq$3 log points)	r = 0.15; -0.18 without Sacramento 2020; median $\lvert\Delta\rvert$ 2.3 vs. 3.0 at other metro-quarters
Washington, DC exclusion	banks and hospital trajectories	banks unchanged ($-18.1/-18.1$); hospitals $-3.4/-3.3$
Philadelphia exclusion	majority-Black bank change, corrected vs. uncorrected	corrected -32.1 to -30.3; uncorrected $+1.3$ to -18.6
Equity cuts, 2009–13 attributes	profile and incidence contrasts	replicate (e.g. banks $-32.3/-17.3$)
Seasonality	quarter-of-year effects ex-2020/21	$\leq$0.9 log points

Philadelphia supplies 84% of the archive-entry correction carried by majority-Black block groups, its SEPTA feeds entering the archive mid-panel. Excluding it, the corrected majority-Black bank change moves from -32.1 to -30.3 percent and the uncorrected change from $+1.3$ to -18.6; majority-white estimates move from -17.1 to -16.9 corrected.

Table A.4: Trajectory endpoints by travel-time budget.

Service	15 min	30 min	45 min	60 min
Banks	−5.6	−12.8	−18.1	−21.2
Libraries	−0.1	−1.0	−2.5	−4.3
Groceries	+1.6	+1.8	+1.7	+1.3
Community colleges	0.0	−0.2	−0.4	−1.0
Hospitals	−0.2	−1.2	−3.4	−5.4

Percent change, 2025 vs. 2014 annual average, the panel, corrected, population-weighted.

Table A.5: Construct grid.

	Coverage: % reaching ≥1 in 45 min			Nearest facility (min)	
Service	2014	2025	Change (pp)	Median 2023	Change
Banks	75.5	72.2	−3.3	28	+2
Libraries	48.8	48.3	−0.5	48	0
Groceries	71.1	71.3	+0.2	29	0
Community colleges	10.0	10.5	+0.5	>90	—
Hospitals	24.2	22.7	−1.5	76	+2

Coverage (share of population reaching at least one facility in 45 minutes, pp change) and nearest-facility (population-weighted minutes to the closest facility) variants, the panel, 2025 vs. 2014. The nearest-facility sweep reads the same travel-time matrices, which hold trips of up to 90 minutes; a median beyond that is reported as >90 with no change, and nearest-facility changes are reported at nearest-minute precision.

Table A.6: Quality flags: classes, detection rules, and coverage.

Flag class	Detection rule	Cells caught	Share
Feed outage or defect	metro-quarter with a partial outage, a degraded or duplicated archived feed, or an archive loss contradicted by agency-reported revenue miles	136 of 2,330 metro-quarters	5.8%
Thin feed	routable feed with implausibly sparse service	0 beyond outages	0.0%
Metro-years touched by any flag		70 of 606	11.6%
Metros ever flagged (excluded from the core)		29	
Never flagged, missing 2014 or 2025 (excluded)		4	
Never flagged, observed 2014–2025 (the core)		19	

Table A.7: Per-exhibit samples, units, and weights.

Exhibit	Sample	Units	Weights
Figs. 1–5 (decompositions)	clean-archive core (19 metros; 28,147 BGs)	log points	population
Trajectories (§5)	the panel (52 balanced metros ex-NYC)	percent	population
Fig. 6 (group bounds)	the panel	percent	population
Fig. 7 (Seattle map)	Seattle metro (2,733 BGs)	loss tiers	—
Table 3 (cross-service)	clean-archive core	log points	population
Table A.8 (two samples)	core + all-metro companion (35 metros; 55,359 BGs; community colleges 34 metros, 55,099 BGs)	log points	population
Compounding (§6.3)	clean-archive core	BG counts	unweighted
NYC section	New York (14,472 BGs; uncorrected)	log points	population
Figs. A.1–A.2 (appendix)	core vs excluded metros; core vs all-metro companion; Cincinnati	log points	population

Trajectories and decompositions reconciled in named steps. The paper quotes two endpoint objects, and for banks they differ (trajectory $-18.1\% = -19.9$ log points; core decomposition endpoint -21.4). Construction and coverage (fixed-effects quarter averages on the panel $\rightarrow$ annual means on the panel's every-year block groups): -19.9 to -17.9, $+2.0$ log points. Archive-quality composition (all-metro companion $\rightarrow$ clean-archive core): -17.9 to -21.4, -3.5 log points. For groceries the same steps run $+1.7 \rightarrow +1.9 \rightarrow +0.4$.

Table A.8: Endpoint decompositions under both samples.

	Clean-archive core				All-metro companion			
Service	Obs.	Net.	Scale	Reloc.	Obs.	Net.	Scale	Reloc.
Bank branches	−21.4	−4.0	−23.6	+6.2	−17.9	−3.2	−19.9	+5.2
Public libraries	−1.0	−2.5	+3.3	−1.8	−1.6	−2.4	+1.9	−1.1
Groceries (full-line)	+0.4	−2.8	+6.5	−3.3	+1.9	−2.7	+8.4	−3.8
Community colleges	0.0	0.0	+3.6	−3.6	−0.3	−0.3	+1.8	−1.8
Durable general acute hospitals	−2.4	−1.5	−1.2	+0.3	−3.1	−1.4	−4.8	+3.0

2025 vs. 2014 annual average, log points. Components sum to observed within rounding. The community-college companion row covers 34 metros (55,099 block groups), because Anchorage has no campus in some years.